Chapter 5. Logistic Regression

Interpretation, inference and fitting

Contents

5.1 Interpreting parameters in logistic regression **2**
- 5.1.1 Interpreting β: odds, probabilities, and linear approximations . . . 2
- 5.1.2 Looking at the data . . . 3
- 5.1.3 Horseshoe crabs revisited . . . 3
- 5.1.4 Logistic regression with retrospective studies . . . 4

5.2 Inference for logistic regression **5**
- 5.2.1 Types of inference . . . 5
- 5.2.2 Inference for horseshoe crab data . . . 6
- 5.2.3 Checking goodness of fit: ungrouped and grouped data . . . 6
- 5.2.4 Goodness of fit of model for horseshoe crabs . . . 6
- 5.2.5 Checking goodness of fit with ungrouped data by grouping . . . 7

5.3 Logit models with categorical predictors **8**
- 5.3.1 ANOVA-type representation of factors . . . 8
- 5.3.2 Dummy variables in logit models . . . 8
- 5.3.3 Alcohol and infant malformation example revisited . . . 9
- 5.3.4 Linear logit model for $I \times 2$ tables . . . 9
- 5.3.5 Cochran–Armitage trend test . . . 9

5.4 Multiple logistic regression **11**
- 5.4.1 Logit models for multiway contingency tables . . . 11
- 5.4.2 AIDS and AZT example . . . 12
- 5.4.3 Goodness of fit as a likelihood-ratio test . . . 12
- 5.4.4 Horseshoe crab example revisited . . . 13
- 5.4.5 Model comparison . . . 13
- 5.4.6 Quantitative treatment of ordinal predictor . . . 13
- 5.4.7 Standardized and probability-based interpretations . . . 14

5.5 Fitting logistic regression models **14**
- 5.5.1 Likelihood equations . . . 14
- 5.5.2 Asymptotic covariance matrix of parameter estimators . . . 15
- 5.5.3 Distribution of probability estimators . . . 15
- 5.5.4 Newton–Raphson method applied to logistic regression . . . 15
- 5.5.5 Convergence and existence of finite estimates . . . 16

Logistic regression is the most important model for categorical responses, used in biomedicine, social science, marketing, credit scoring and genetics. Section 5.1 interprets its parameters, Section 5.2 presents inference, Sections 5.3–5.4 extend to several and to qualitative predictors, and Section 5.5 applies the GLM theory of Chapter 4 to the likelihood equations and their solution.

5.1 Interpreting parameters in logistic regression

Motivation. The logistic model is linear on the logit scale, but people think in probabilities. We need several readings of β: an odds ratio, a local slope $\beta\pi(1-\pi)$ for the probability, and changes in π over the range of x.

For binary Y and predictor X let $\pi(x) = P(Y = 1 \mid X = x)$. The logistic regression model is

$$\pi(x) = \frac{\exp(\alpha+\beta x)}{1+\exp(\alpha+\beta x)}, \qquad \text{equivalently} \qquad \operatorname{logit}\pi(x) = \log\frac{\pi(x)}{1-\pi(x)} = \alpha+\beta x. \tag{5.1}$$

5.1.1 Interpreting β: odds, probabilities, and linear approximations

The sign of β says whether $\pi(x)$ increases or decreases; the curve gets steeper as $|\beta|$ grows and flattens to a constant as $\beta \to 0$, where Y and X are independent. For $\beta > 0$ the curve is a logistic cdf; the symmetry of the logistic density means $\pi(x)$ approaches 1 at the same rate as it approaches 0.

Proposition 5.1 (Odds-ratio interpretation)

Under (5.1) the odds are $\pi(x)/[1-\pi(x)] = e^{\alpha}(e^{\beta})^{x}$; hence for every x,

$$\frac{\pi(x+1)/[1-\pi(x+1)]}{\pi(x)/[1-\pi(x)]} = e^{\beta}:$$

the odds multiply by e^{β} for each unit increase in x, and e^{β} is an odds ratio.

Proof. Exponentiate the logit form: odds $= \exp(\alpha+\beta x)$. The ratio at $x+1$ and x is $\exp(\alpha+\beta x+\beta)/\exp(\alpha+\beta x) = e^{\beta}$. ∎

Proposition 5.2 (Linear approximation (Berkson 1951))

Under (5.1),

$$\frac{d\pi(x)}{dx} = \beta\,\pi(x)[1-\pi(x)].$$

(a) The slope is greatest, $\beta/4$, where $\pi(x) = \frac{1}{2}$, i.e. at $x = -\alpha/\beta$ (the median effective level EL_{50}; in toxicology the LD_{50}). At $\pi(x) = 0.9$ or 0.1 it is 0.09β, and it tends to 0 as $\pi(x) \to 0$ or 1.
(b) $\pi(-\alpha/\beta \pm 1/\beta) = e^{\pm 1}/(1+e^{\pm 1})$, i.e. 0.73 and 0.27: a change of $1/\beta$ in x from the EL_{50} changes π by about 0.23, close to the value $\frac{1}{4}$ suggested by the tangent line.

Proof. [Supplement]

Step 1. With $\eta = \alpha+\beta x$ and $\pi = e^{\eta}/(1+e^{\eta})$,

$$\frac{d\pi}{d\eta} = \frac{e^{\eta}(1+e^{\eta}) - e^{\eta}e^{\eta}}{(1+e^{\eta})^2} = \frac{e^{\eta}}{(1+e^{\eta})^2} = \pi(1-\pi),$$

and $d\eta/dx = \beta$, so $d\pi/dx = \beta\pi(1-\pi)$.

Step 2. The function $\pi(1-\pi)$ on $(0,1)$ is maximized at $\pi = \frac{1}{2}$ with value $\frac{1}{4}$; $\pi(x) = \frac{1}{2}$ iff $\alpha+\beta x = 0$ iff $x = -\alpha/\beta$. Also $0.9 \times 0.1 = 0.09$.

Step 3. At $x = -\alpha/\beta \pm 1/\beta$, $\eta = \pm 1$, so $\pi = e^{\pm 1}/(1+e^{\pm 1}) = 0.731$ or 0.269. ■

So near a point with probability π, a unit change in x changes the probability by roughly $\beta\pi(1-\pi)$ (Figure 1); the approximation is better for smaller changes. Another summary reports $\pi(x)$ at the quartiles of x; the change over the middle half of the data can be compared across predictors with different units. The intercept is rarely of interest, but after centering ($x \mapsto x - \bar{x}$) it is the logit at $\bar{x}$, and centering reduces correlations among estimates in models with quadratic or interaction terms.

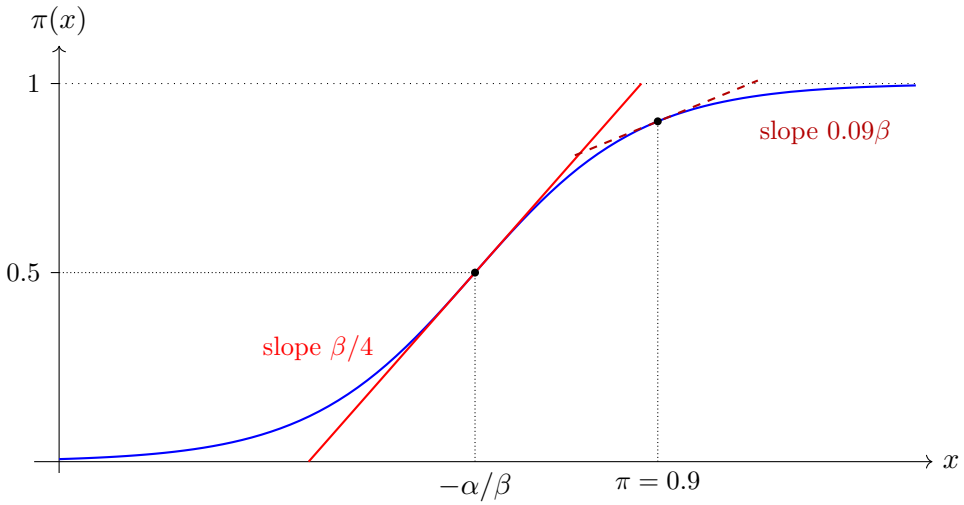


Figure 1: Logistic curve ($\alpha = -3$, $\beta = 1$) with tangent lines: slope $\beta\pi(1-\pi)$ equals $\beta/4$ at $\pi = \frac{1}{2}$ and 0.09β at $\pi = 0.9$.

5.1.2 Looking at the data

Before fitting, check that a logistic curve is plausible. Plotting 0/1 responses against x is uninformative; instead plot sample proportions or sample logits. With y_i successes out of n_i at setting i, the sample logit $\log[y_i/(n_i - y_i)]$ is infinite when $y_i = 0$ or n_i; the adjusted version

$$\log \frac{y_i + \frac{1}{2}}{n_i - y_i + \frac{1}{2}}$$

is the least-biased estimator of this form (bias of order n_i^{-2}, Haldane 1956). The plot should be roughly linear. With continuous x and small n_i, group nearby x values or, better, smooth the data and look for marked departures from an S-shape.

5.1.3 Horseshoe crabs revisited

Example 5.3 (Horseshoe crabs: width and presence of satellites)

For the 173 crabs of Example 4.5, let $Y = 1$ if a female has at least one satellite. The GAM fit and the sample proportions in the eight width categories suggest an increasing S-shaped trend. The ML fit with $x =$ width (cm) is

$$\hat{\pi}(x) = \frac{\exp(-12.351 + 0.497x)}{1 + \exp(-12.351 + 0.497x)}.$$

Parameter	Estimate	SE	LR 95% limits	Wald χ^2
Intercept	-12.3508	2.6287	$(-17.8097, -7.4573)$	22.07
width	0.4972	0.1017	(0.3084, 0.7090)	23.89

Deviance 194.45 and Pearson $X^2 = 165.14$ on 171 df; log likelihood -97.23.

(a) At the mean width 26.3, $\hat\pi = 0.674$; $\hat\pi = \frac{1}{2}$ at $x = 12.351/0.497 = 24.8$.
(b) The odds multiply by $e^{0.497} = 1.64$ per cm (a 64% increase).
(c) At $\bar x$ the probability rises by about $0.497(0.674)(0.326) = 0.11$ per cm.
(d) At the width quartiles 24.9, 26.1, 27.7, $\hat\pi = 0.51,\ 0.65,\ 0.81$: an increase of 0.30 over the middle half.
(e) With weight (kg) instead, $\operatorname{logit}\hat\pi = -3.695 + 1.815x$; $\hat\beta$'s are not comparable across units, but at the weight quartiles 2.00, 2.35, 2.85 we get $\hat\pi = 0.48,\ 0.64,\ 0.81$, an increase of 0.33, similar to width.

5.1.4 Logistic regression with retrospective studies

In case–control studies X is observed for samples of cases ($Y = 1$) and controls ($Y = 0$), so Y is fixed by design. Effects in logistic regression are odds ratios, and odds ratios can be estimated retrospectively.

Theorem 5.4 (Logistic regression under case–control sampling)

Let $Z = 1$ indicate that a subject is sampled, and suppose the sampling probabilities depend only on y: $P(Z = 1 \mid y, x) = \rho_y$ for $y = 0, 1$. If $\operatorname{logit} P(Y = 1 \mid x) = \alpha + \beta x$, then

$$\operatorname{logit} P(Y = 1 \mid Z = 1, x) = \alpha^* + \beta x, \qquad \alpha^* = \alpha + \log(\rho_1/\rho_0).$$

So the slope β is the same as in the prospective model; only the intercept changes.

Proof.

Step 1. By Bayes' theorem,

$$P(Y = 1 \mid Z = 1, x) = \frac{P(Z = 1 \mid Y = 1, x)P(Y = 1 \mid x)}{\sum_{j=0}^{1} P(Z = 1 \mid Y = j, x)P(Y = j \mid x)} = \frac{\rho_1 P(Y = 1 \mid x)}{\rho_0 P(Y = 0 \mid x) + \rho_1 P(Y = 1 \mid x)}.$$

Step 2. Divide numerator and denominator by $P(Y = 0 \mid x)$ and use $P(Y = 1 \mid x)/P(Y = 0 \mid x) = \exp(\alpha + \beta x)$:

$$P(Y = 1 \mid Z = 1, x) = \frac{\rho_1 \exp(\alpha + \beta x)}{\rho_0 + \rho_1 \exp(\alpha + \beta x)} = \frac{\exp[\alpha + \log(\rho_1/\rho_0) + \beta x]}{1 + \exp[\alpha + \log(\rho_1/\rho_0) + \beta x]}.$$

■

If cases are sampled at 10 times the rate of controls, the estimated intercept is $\log 10 = 2.3$ larger than prospectively. For other binary links the effect for Y given X is not recoverable from case–control data; this is a major reason for the popularity of logit models in biomedicine. Matched case–control studies need conditional methods (Section 10.2.5).

Proposition 5.5 (Normal discriminant model implies logistic regression (Cornfield 1962))

Suppose $X \mid Y = i \sim N(\mu_i, \sigma^2)$, $i = 0, 1$, and $P(Y = 1) = \rho$. Then $P(Y = 1 \mid X = x)$ satisfies (5.1) with

$$\beta = \frac{\mu_1 - \mu_0}{\sigma^2}, \qquad \alpha = \log\frac{\rho}{1-\rho} - \frac{\mu_1^2 - \mu_0^2}{2\sigma^2}.$$

If the two variances differ, the logit is quadratic in x.

Proof. [Supplement] Let φ_i be the $N(\mu_i, \sigma_i^2)$ density. By Bayes' theorem,

$$\operatorname{logit} P(Y = 1 \mid x) = \log\frac{\rho\varphi_1(x)}{(1-\rho)\varphi_0(x)} = \log\frac{\rho}{1-\rho} + \log\frac{\sigma_0}{\sigma_1} - \frac{(x-\mu_1)^2}{2\sigma_1^2} + \frac{(x-\mu_0)^2}{2\sigma_0^2}.$$

With $\sigma_0 = \sigma_1 = \sigma$ the x^2 terms cancel and

$$-\frac{(x-\mu_1)^2 - (x-\mu_0)^2}{2\sigma^2} = \frac{(\mu_1 - \mu_0)x}{\sigma^2} - \frac{\mu_1^2 - \mu_0^2}{2\sigma^2},$$

giving the stated α, β. If $\sigma_0 \neq \sigma_1$ the coefficient of x^2 is $\frac{1}{2}(\sigma_0^{-2} - \sigma_1^{-2}) \neq 0$. ■

So when a population mixes two roughly normal groups with similar spread, the logistic curve fits well; with unequal spreads $\pi(x)$ is nonmonotone (rises then falls, or the reverse).

5.2 Inference for logistic regression

Motivation. ML estimators in logistic regression are asymptotically normal, so the Wald, likelihood-ratio and score triad of Theorem 4.18 applies. The likelihood-ratio versions are generally the most reliable, and goodness of fit needs care when predictors are continuous.

5.2.1 Types of inference

For $\operatorname{logit}\pi(x) = \alpha + \beta x$, the test of independence $H_0 : \beta = 0$ can use

(i) Wald: $z = \hat\beta/\widehat{\mathrm{se}}$, or $z^2 \sim \chi_1^2$;

(ii) likelihood ratio: $-2(L_0 - L_1)$, twice the difference of maximized log likelihoods under H_0 and the full model, χ_1^2;

(iii) score: based on the derivative of the log likelihood at $\beta = 0$, comparing the sufficient statistic for β with its null expectation (Section 5.3.5).

In large samples they agree. The LR test uses the log likelihood at both H_0 and $\hat\beta$ and is preferred; for large $|\beta|$ the Wald test loses power and can behave aberrantly (Hauck and Donner 1977). Inverting a test gives a confidence interval: Wald gives $\hat\beta \pm z_{\alpha/2}\widehat{\mathrm{se}}$; the LR inversion is the profile-likelihood interval.

Proposition 5.6 (Interval for $\pi(x_0)$)

The estimated variance of the fitted logit at x_0 is

$$\widehat{\operatorname{Var}}(\hat\alpha + \hat\beta x_0) = \widehat{\operatorname{Var}}(\hat\alpha) + x_0^2\,\widehat{\operatorname{Var}}(\hat\beta) + 2x_0\,\widehat{\operatorname{Cov}}(\hat\alpha, \hat\beta),$$

and if (l, u) is a $1-\alpha$ interval for $\operatorname{logit}\pi(x_0)$, then $(e^l/(1+e^l),\ e^u/(1+e^u))$ is a $1-\alpha$ interval for $\pi(x_0)$.

Proof. The first formula is the variance of the linear combination $(1, x_0)(\hat{\alpha}, \hat{\beta})^\top$, namely $(1, x_0)\widehat{\mathrm{Cov}}(\hat{\alpha}, \hat{\beta})(1, x_0)^\top$. Since $t \mapsto e^t/(1+e^t)$ is strictly increasing, $l \le \operatorname{logit}\pi(x_0) \le u$ iff $e^l/(1+e^l) \le \pi(x_0) \le e^u/(1+e^u)$, so the coverage is preserved. ■

Equivalently, refit with $x^* = x - x_0$: the intercept and its SE are the fitted logit at x_0 and its SE.

5.2.2 Inference for horseshoe crab data

Example 5.7 (Horseshoe crabs: tests and intervals)

(a) Wald: $z = 0.497/0.102 = 4.9$, $z^2 = 23.9$ ($P < 0.0001$). The maximized log likelihoods are -112.88 under $\beta = 0$ and -97.23 for the model, so the LR statistic is $-2(-112.88 + 97.23) = 31.3$ (df $= 1$), even stronger evidence.
(b) The Wald interval for β is $0.497 \pm 1.96(0.102) = (0.298, 0.697)$; the profile-likelihood interval is (0.308, 0.709). For the odds ratio per cm: $(e^{0.308}, e^{0.709}) = (1.36, 2.03)$: at least a 36% increase and at most a doubling of the odds.
(c) At $x = 26.5$: logit $= -12.351 + 0.497(26.5) = 0.825$, $\hat{\pi} = 0.695$. With $\widehat{\mathrm{Var}}(\hat{\alpha}) = 6.910$, $\widehat{\mathrm{Var}}(\hat{\beta}) = 0.01035$, $\widehat{\mathrm{Cov}} = -0.2668$,

$$\widehat{\mathrm{Var}} = 6.910 + (26.5)^2(0.01035) + 2(26.5)(-0.2668) = 0.038,$$

so the logit interval is $0.825 \pm 1.96\sqrt{0.038} = (0.44, 1.21)$, i.e. (0.61, 0.77) for $\pi(26.5)$.
(d) Only 6 crabs had $x = 26.5$, 4 with satellites: the sample proportion 0.67 has SE $= \sqrt{0.67(0.33)/6} = 0.19$ and score interval (0.30, 0.90), versus model SE $= 0.04$. The model borrows strength from all 173 observations to estimate two parameters; when it approximates the truth decently, its estimates are much more precise (Section 6.4.5).

5.2.3 Checking goodness of fit: ungrouped and grouped data

One way to detect lack of fit compares the model with a more complex one (quadratic term, interaction) by an LR test; if the extra terms do not help, the model gains support. Global tests compare observed counts with fitted values $n_i\hat{\pi}_i$, $n_i(1-\hat{\pi}_i)$ through X^2 or G^2; for a fixed number of predictor settings, as fitted counts grow, they are χ^2 with residual df = (number of settings) − (number of parameters). This requires categorical predictors: as noted in Section 4.5.3, the saturated model differs for grouped and ungrouped data, and the chi-squared limit needs a fixed number of settings.

5.2.4 Goodness of fit of model for horseshoe crabs

Example 5.8 (Horseshoe crabs: goodness of fit)

(a) Adding a quadratic term with centered width gives logit $\hat{\pi} = 0.618 + 0.533x + 0.040x^2$, with $\widehat{\mathrm{se}} = 0.046$ for the quadratic coefficient; the LR statistic is 0.83 (df $= 1$): no need for it.
(b) Width takes 66 distinct values; treating the data as a 66×2 table violates the chi-squared theory (tiny fitted counts, and more data would add cells). Grouping into the eight width categories of Example 4.5, with fitted values the sums of $\hat{\pi}(x)$ and $1 - \hat{\pi}(x)$ over the crabs in each category:

Width (cm)	Yes	No	Fitted yes	Fitted no
< 23.25	5	9	3.64	10.36
23.25–24.25	4	10	5.31	8.69
24.25–25.25	17	11	13.78	14.22
25.25–26.25	21	18	24.23	14.77
26.25–27.25	15	7	15.94	6.06
27.25–28.25	20	4	19.38	4.62
28.25–29.25	15	3	15.65	2.35
> 29.25	14	0	13.08	0.92

Then $X^2 = 5.3$ and $G^2 = 6.2$ with df $= 8 - 2 = 6$ ($P > 0.4$): no evidence of lack of fit (the theory is still approximate since $\pi(x)$ varies within a category).

5.2.5 Checking goodness of fit with ungrouped data by grouping

With many predictors, grouping on each variable gives a sparse table. Instead, partition the observations by their fitted probabilities.

Definition (Hosmer–Lemeshow statistic)

Order the observations by $\hat{\pi}$ from the ungrouped fit and form g groups of roughly equal size (usually $g = 10$). With y_{ij} the binary outcome and $\hat{\pi}_{ij}$ the fitted probability of observation j in group i ($j = 1, \ldots, n_i$),

$$\mathrm{HL} = \sum_{i=1}^{g} \frac{\left(\sum_j y_{ij} - \sum_j \hat{\pi}_{ij}\right)^2}{\left(\sum_j \hat{\pi}_{ij}\right)\left[1 - \left(\sum_j \hat{\pi}_{ij}\right)/n_i\right]}.$$

Proposition 5.9 (HL as a Pearson statistic)

HL equals the Pearson statistic $\sum(\text{observed} - \text{fitted})^2/\text{fitted}$ for the $g \times 2$ table of observed and fitted counts of successes and failures in the groups.

Proof. [Supplement] In group i let $O_i = \sum_j y_{ij}$, $E_i = \sum_j \hat{\pi}_{ij}$. The failure cell has observed $n_i - O_i$ and fitted $n_i - E_i$, so its deviation is $-(O_i - E_i)$. The two Pearson terms of row i add to

$$(O_i - E_i)^2\Big(\frac{1}{E_i} + \frac{1}{n_i - E_i}\Big) = \frac{(O_i - E_i)^2\, n_i}{E_i(n_i - E_i)} = \frac{(O_i - E_i)^2}{E_i(1 - E_i/n_i)}.$$

Summing over i gives HL. ■

Reference distribution and limitations

HL is not asymptotically chi-squared in the usual sense, since observations within a group do not share a success probability; Hosmer and Lemeshow found by simulation that when the number of distinct covariate patterns equals n, the null distribution is approximately χ^2_{g-2}. Ties in $\hat{\pi}$ make the grouping somewhat arbitrary, so software values differ slightly. For the crab width model with $g = 10$, HL $= 3.5$ on 8 df: a decent fit. Like other global statistics, HL has low power against specific alternatives and does not say what is wrong; comparing with a more complex model is more informative, and diagnostics (Section 6.2) locate problems.

5.3 Logit models with categorical predictors

Motivation. Qualitative predictors enter through dummy variables. Different codings change the individual parameters but not the fit or the odds ratios; for ordered factors, scores lead to the linear logit model and the Cochran–Armitage trend test.

5.3.1 ANOVA-type representation of factors

For a single factor X with I categories, let $y_i \sim \text{bin}(n_i, \pi_i)$ in row i of an $I \times 2$ table. The model

$$\log \frac{\pi_i}{1-\pi_i} = \alpha + \beta_i, \qquad i = 1, \ldots, I, \tag{5.2}$$

resembles one-way ANOVA and has one redundant parameter, so a constraint is imposed, e.g. $\beta_I = 0$.

Proposition 5.10 (Coding invariance and saturation)

(a) Model (5.2) is saturated: for any $\{\pi_i\} \subset (0,1)$ there are parameters satisfying it, and its fitted values are the sample proportions.
(b) Under any constraint, $\alpha + \beta_i = \operatorname{logit} \pi_i$ and $\beta_a - \beta_b$ is the log odds ratio for rows a, b; these and the fit are the same for every coding. With $\beta_I = 0$, $\alpha = \operatorname{logit} \pi_I$ and β_i is the log odds ratio of row i versus row I.
(c) No effect, $\beta_1 = \cdots = \beta_I$, is statistical independence of X and Y.

Proof.

(a) Take $\alpha = \operatorname{logit} \pi_I$ and $\beta_i = \operatorname{logit} \pi_i - \operatorname{logit} \pi_I$. The model has I free parameters for I binomials, so it is saturated and $\hat{\pi}_i = y_i/n_i$.

(b) If $(\alpha, \boldsymbol{\beta})$ satisfies the model, so does $(\alpha - c, \beta_i + c)$ for any c, and every constraint selects one such c; $\alpha + \beta_i$ and $\beta_a - \beta_b$ do not depend on c. In particular $\beta_a - \beta_b = \operatorname{logit} \pi_a - \operatorname{logit} \pi_b$, a log odds ratio.

(c) Equal β_i means equal π_i across rows. ■

5.3.2 Dummy variables in logit models

Equivalently, with dummy variables $x_i = 1$ for row i and 0 otherwise ($i = 1, \ldots, I-1$),

$$\operatorname{logit} \pi = \alpha + \beta_1 x_1 + \cdots + \beta_{I-1} x_{I-1};$$

no dummy for category I corresponds to $\beta_I = 0$ (some software instead sets $\beta_1 = 0$). *Effect coding* imposes $\sum_i \beta_i = 0$; for $I = 2$ it uses $x = 1$ in category 1 and $x = -1$ in category 2, and then the log odds ratio is $\beta_1 - \beta_2 = 2\beta_1$, whereas with a 0/1 dummy it is β_1. A single $\hat{\beta}_i$ is meaningful only in comparison with another category.

5.3.3 Alcohol and infant malformation example revisited

Example 5.11 (Maternal alcohol consumption and malformation)

Alcohol (drinks/day)	Present	Absent	Logit	Observed prop.	Fitted (linear logit)
0	48	17066	−5.87	0.0028	0.0026
< 1	38	14464	−5.94	0.0026	0.0030
1–2	5	788	−5.06	0.0063	0.0041
3–5	1	126	−4.84	0.0079	0.0091
≥ 6	1	37	−3.61	0.0263	0.0231

For (5.2) with malformation as response, $\hat{\alpha} + \hat{\beta}_i$ are the sample logits, e.g. $\log(48/17066) = -5.87$. With $\beta_5 = 0$: $\hat{\alpha} = -3.61$, $\hat{\beta}_1 = -2.26$; with $\beta_1 = 0$: $\hat{\alpha} = -5.87$. Apart from a slight reversal in the first two rows, the logits increase with alcohol. Independence has $\hat{\alpha} = \log(93/32481) = -5.86$; testing it (df $= 4$) gives $X^2 = 12.1$ ($P = 0.02$) but $G^2 = 6.2$ ($P = 0.19$): mixed signals, since the counts are very unbalanced (exact conditional P-values 0.03 and 0.13). Both ignore the ordering of alcohol, and one case more or less in the last rows changes those proportions substantially.

5.3.4 Linear logit model for $I \times 2$ tables

For ordered categories with scores $x_1, \ldots, x_I$ the *linear logit model*

$$\operatorname{logit} \pi_i = \alpha + \beta x_i \tag{5.3}$$

lies between independence ($\beta = 0$) and the saturated model (5.2).

Example 5.12 (Alcohol: linear logit fit)

With scores (0, 0.5, 1.5, 4.0, 7.0) (the last somewhat arbitrary): $\hat{\alpha} = -5.9605$ (0.1154) and $\hat{\beta} = 0.3166$ (0.1254), profile-likelihood interval (0.0187, 0.5236). The estimated odds of malformation multiply by $e^{0.317} = 1.37$ per drink per day. Fit: $G^2 = 1.95$, $X^2 = 2.05$, df $= 3$; log likelihood -635.60. The fitted proportions are in the table of Example 5.11.

5.3.5 Cochran–Armitage trend test

For an $I \times 2$ table with ordered rows and independent $y_i \sim \text{bin}(n_i, \pi_i)$, Cochran (1954) and Armitage (1955) partitioned the Pearson statistic using the linear probability model $\pi_i = \alpha + \beta x_i$ fitted by (weighted) least squares. Let $n = \sum n_i$, $p_i = y_i/n_i$, $p = \sum y_i/n$ and $\bar{x} = \sum n_i x_i/n$. The prediction equation is $\hat{\pi}_i = p + b(x_i - \bar{x})$ with

$$b = \frac{\sum_i n_i (p_i - p)(x_i - \bar{x})}{\sum_i n_i (x_i - \bar{x})^2}.$$

Theorem 5.13 (Partition of the Pearson statistic)

Let $X^2(I) = \sum_i n_i (p_i - p)^2 / [p(1-p)]$ be the Pearson statistic for independence in the $I \times 2$ table. Then

$$X^2(I) = z^2 + X^2(L),$$

where

$$X^2(L) = \frac{\sum_i n_i(p_i - \hat{\pi}_i)^2}{p(1-p)}, \qquad z^2 = \frac{b^2 \sum_i n_i(x_i - \bar{x})^2}{p(1-p)} = \frac{\left[\sum_i (x_i - \bar{x}) y_i\right]^2}{p(1-p) \sum_i n_i (x_i - \bar{x})^2}.$$

Under independence z^2 is asymptotically χ^2_1 (the *Cochran–Armitage trend test* of $\beta = 0$); when the linear probability model holds, $X^2(L)$ is asymptotically χ^2_{I-2} (a test of linearity).

Proof. [Supplement]

Step 1. (The Pearson form.) The cells of row i have observed y_i, $n_i - y_i$ and fitted $n_i p$, $n_i(1-p)$, with deviations $\pm(y_i - n_i p)$. As in Proposition 5.9 the two terms add to $(y_i - n_i p)^2/[n_i p(1-p)] = n_i(p_i - p)^2/[p(1-p)]$.

Step 2. (Decomposition.) Write $p_i - p = (p_i - \hat{\pi}_i) + (\hat{\pi}_i - p)$. Then

$$\sum n_i (p_i - p)^2 = \sum n_i (p_i - \hat{\pi}_i)^2 + \sum n_i (\hat{\pi}_i - p)^2 + 2 \sum n_i (p_i - \hat{\pi}_i)(\hat{\pi}_i - p).$$

Step 3. (Cross term vanishes.) Since $\hat{\pi}_i - p = b(x_i - \bar{x})$,

$$\sum n_i (p_i - \hat{\pi}_i)(x_i - \bar{x}) = \sum n_i (p_i - p)(x_i - \bar{x}) - b \sum n_i (x_i - \bar{x})^2 = 0$$

by the definition of b.

Step 4. (Middle term.) $\sum n_i(\hat{\pi}_i - p)^2 = b^2 \sum n_i (x_i - \bar{x})^2$. Since $\sum n_i (x_i - \bar{x}) = 0$, $\sum n_i (p_i - p)(x_i - \bar{x}) = \sum (y_i - n_i p)(x_i - \bar{x}) = \sum (x_i - \bar{x}) y_i$, which gives the second form of z^2. Dividing Step 2 by $p(1-p)$ yields the partition.

Step 5. (Null distribution of z^2.) Under independence $E \sum (x_i - \bar{x}) Y_i = \pi \sum n_i (x_i - \bar{x}) = 0$ and $\mathrm{Var} \sum (x_i - \bar{x}) Y_i = \pi(1-\pi) \sum n_i (x_i - \bar{x})^2$. By the CLT and $p \xrightarrow{p} \pi$ (Slutsky), $z \xrightarrow{d} N(0,1)$. The limit of $X^2(L)$ follows from the chi-squared theory for the fit of a 2-parameter model to I binomials. ■

Proposition 5.14 (Cochran–Armitage = score test)

The score statistic for $H_0 : \beta = 0$ in the linear logit model (5.3) equals z^2 of Theorem 5.13.

Proof. [Supplement]

Step 1. Under H_0 the MLE is $\tilde{\alpha} = \mathrm{logit}\, p$, so $\tilde{\pi}_i = p$. By (4.16) in count form the score components at $(\tilde{\alpha}, 0)$ are

$$u_\alpha = \sum_i (y_i - n_i p) = 0, \qquad u_\beta = \sum_i x_i (y_i - n_i p) = \sum_i (x_i - \bar{x}) y_i.$$

Step 2. The information at $(\tilde{\alpha}, 0)$ is $X^\top \mathrm{diag}\{n_i p(1-p)\} X$:

$$\mathcal{I} = p(1-p) \begin{pmatrix} n & \sum n_i x_i \\ \sum n_i x_i & \sum n_i x_i^2 \end{pmatrix},$$

and by the formula for the inverse of a 2×2 matrix,

$$[\mathcal{I}^{-1}]_{\beta\beta} = \frac{n}{p(1-p)\left[n \sum n_i x_i^2 - (\sum n_i x_i)^2\right]} = \frac{1}{p(1-p) \sum n_i (x_i - \bar{x})^2}.$$

Step 3. The score statistic $\boldsymbol{u}^\top \mathcal{I}^{-1} \boldsymbol{u}$ with $u_\alpha = 0$ is $u_\beta^2 [\mathcal{I}^{-1}]_{\beta\beta} = z^2$. ■

The statistic also equals $M^2 = (n-1) r^2$ of Section 3.4 for $J = 2$, with $n - 1$ replaced by n; for $I = 2$, $X^2(L) = 0$ and $z^2 = X^2(I)$.

Example 5.15 (Alcohol: trend test)

With the scores above, $z^2 = 6.6$ ($P = 0.010$), and $X^2(I) = 12.1 = 6.6 + 5.5$, where $X^2(L) = 5.5$ (df $= 3$) shows only slight departure from linearity. The correlation is only $r = 0.014$ ($n = 32{,}574$), but r is a poor descriptive measure for such unbalanced tables. The Wald statistic is $(0.317/0.125)^2 = 6.4$ ($P = 0.012$) and the LR statistic 4.25 ($P = 0.039$); given the highly unbalanced counts the LR approach is safest, and the profile interval (0.02, 0.52) is preferable to the Wald interval (0.07, 0.56). Exact methods (Section 6.7.4) are relevant even though n is huge.

5.4 Multiple logistic regression

Motivation. With several predictors each coefficient is a conditional effect, controlling for the others. Comparing nested models by deviance differences is the main tool for deciding which predictors and interactions are needed.

For predictors $\mathbf{x} = (x_1, \ldots, x_p)$,

$$\operatorname{logit} \pi(\mathbf{x}) = \alpha + \beta_1 x_1 + \cdots + \beta_p x_p, \qquad \pi(\mathbf{x}) = \frac{\exp(\alpha + \beta_1 x_1 + \cdots + \beta_p x_p)}{1 + \exp(\alpha + \beta_1 x_1 + \cdots + \beta_p x_p)}. \tag{5.4}$$

e^{β_j} is the multiplicative effect on the odds of a unit increase in x_j, at fixed levels of the other predictors.

5.4.1 Logit models for multiway contingency tables

With binary X and Z coded by dummies ($x_1 = z_1 = 1$, $x_2 = z_2 = 0$) and independent binomials at each (i, k), consider

$$\operatorname{logit} P(Y = 1) = \alpha + \beta_1 x_i + \beta_2 z_k. \tag{5.5}$$

Proposition 5.16 (Main-effects model and homogeneous association)

In (5.5), at each level z_k of Z,

$$\operatorname{logit} P(Y = 1 \mid x = 1, z_k) - \operatorname{logit} P(Y = 1 \mid x = 0, z_k) = \beta_1,$$

so e^{β_1} is the conditional XY odds ratio, the same at each level of Z (homogeneous association). $\beta_1 = 0$ is conditional independence of X and Y given Z.

Proof. $(\alpha + \beta_1 + \beta_2 z_k) - (\alpha + \beta_2 z_k) = \beta_1$, free of z_k; a difference of logits is a log odds ratio. If $\beta_1 = 0$ the conditional odds ratios all equal 1. ■

No interaction thus means additivity on the logit scale; on another scale (probit, identity) interaction may appear or disappear, and in some applications another scale is natural. An ANOVA-type form handles factors with any numbers of categories:

$$\operatorname{logit} P(Y = 1) = \alpha + \beta_i^X + \beta_k^Z, \tag{5.6}$$

with one redundant parameter per factor (e.g. $\beta_I^X = \beta_K^Z = 0$; superscripts are labels). Conditional independence of X and Y is $\beta_1^X = \cdots = \beta_I^X$. For two categories each, (5.6) with $\beta_2^X = \beta_2^Z = 0$ is (5.5) with $\beta_1^X = \beta_1$, $\beta_1^Z = \beta_2$.

5.4.2 AIDS and AZT example

Example 5.17 (AIDS symptoms, AZT use and race)

338 veterans with faltering immune systems were randomized to immediate AZT or to AZT only after severe T-cell weakness; the response is development of AIDS symptoms over 3 years.

		Symptoms	
Race	AZT use	Yes	No
White	Yes	14	93
	No	32	81
Black	Yes	11	52
	No	12	43

In (5.5) let $x = 1$ for immediate AZT and $z = 1$ for whites. Fit:

Parameter	Estimate	SE	Wald χ^2
Intercept	-1.0736	0.2629	16.67
AZT	-0.7195	0.2790	6.65
race	0.0555	0.2886	0.04

(a) The estimated conditional odds ratio for AZT is $e^{-0.7195} = 0.487$: for each race the odds of symptoms are about halved by immediate AZT. Wald interval $\exp[-0.720 \pm 1.96(0.279)] = (0.28, 0.84)$; profile interval (0.279, 0.835). Race: 1.057, interval (0.60, 1.86).
(b) $H_0 : \beta_1 = 0$ (conditional independence of AZT and symptoms given race): LR statistic 6.9 (df $= 1$, $P = 0.01$); Wald $(-0.720/0.279)^2 = 6.65$.
(c) Codings: for (last $= 0$, first $= 0$, sum $= 0$) the intercept is $(-1.074, -1.738, -1.406)$, AZT-yes $(-0.720, 0, -0.360)$, white $(0.055, 0, 0.028)$. In each, intercept + AZT-yes + white $= -1.738$, so $\hat{\pi} = e^{-1.738}/(1+e^{-1.738}) = 0.15$ for whites with immediate AZT, and $\hat{\beta}_1^X - \hat{\beta}_2^X = -0.720$ always.
(d) Fitted probabilities (95% limits): white/AZT 0.150 (0.099, 0.220); white/no 0.265 (0.197, 0.348); black/AZT 0.143 (0.087, 0.225); black/no 0.255 (0.170, 0.364).

5.4.3 Goodness of fit as a likelihood-ratio test

Write $G^2(M_0 \mid M_1) = -2(L_0 - L_1)$ for the LR statistic testing M_0 within M_1. The goodness-of-fit statistic $G^2(M)$ is the case $M_0 = M$, $M_1 =$ saturated, with df = (number of binomials) − (number of parameters). By Proposition 4.22, with L_S the saturated log likelihood,

$$G^2(M_0 \mid M_1) = -2(L_0 - L_S) - [-2(L_1 - L_S)] = G^2(M_0) - G^2(M_1).$$

Example 5.18 (AIDS: goodness of fit)

Whites with immediate AZT have $\hat{\pi} = 0.150$, so the fitted counts are $107(0.150) = 16.0$ and $107(0.850) = 91.0$; similarly for all 8 cells. Then $G^2 = 1.38$, $X^2 = 1.39$, with df $= 4 - 3 = 1$ ($P > 0.2$). Since the model has no interaction, this is also a test of homogeneous odds ratios; adding the interaction gives the saturated model. For H_0: no race effect, $G^2(M_0) - G^2(M_1) = 1.42 - 1.38 = 0.04$.

The comparison statistic is often nearly chi-squared even when the separate $G^2(M_i)$ are not (continuous predictors, small fitted values), provided the models differ by few parameters.

5.4.4 Horseshoe crab example revisited

Example 5.19 (Horseshoe crabs: width and color)

Color has four observed categories (medium light, medium, medium dark, dark), a surrogate for age. With dummies c_1, c_2, c_3 for the first three (dark is the reference),

$$\operatorname{logit}\pi = \alpha + \beta_1 c_1 + \beta_2 c_2 + \beta_3 c_3 + \beta_4 x. \tag{5.7}$$

Parameter	Estimate	SE	LR 95% limits	χ^2
intercept	−12.7151	2.7618	(−18.4564, −7.5788)	21.20
c_1	1.3299	0.8525	(−0.2738, 3.1354)	2.43
c_2	1.4023	0.5484	(0.3527, 2.5260)	6.54
c_3	1.1061	0.5921	(−0.0279, 2.3138)	3.49
width	0.4680	0.1055	(0.2713, 0.6870)	19.66

Deviance 187.46, Pearson 168.66 on 168 df; log likelihood −93.73.

For dark crabs $\operatorname{logit}\hat\pi = -12.715 + 0.468x$; for medium-light, $-11.385 + 0.468x$. At $x = 26.3$, $\hat\pi = 0.399$ (dark) and 0.715 (medium light). Without interaction the curves for the colors are horizontal shifts of each other and never cross; a 1-cm increase multiplies the odds by $e^{0.468} = 1.60$ for every color. Medium-light versus dark: $e^{1.330} = 3.8$ at any width (at 26.3: $2.51/0.66 = 3.8$).

5.4.5 Model comparison

Example 5.20 (Horseshoe crabs: testing color and interaction)

$H_0 : \beta_1 = \beta_2 = \beta_3 = 0$ (no color effect given width): $-2(L_0 - L_1) = 7.0$, df = 3, $P = 0.07$: slight evidence of a color effect. Adding the three color×width products (equivalent to separate logistic fits per color, so color comparisons depend on width) gives LR = 4.4, df = 3, $P = 0.22$: weak evidence of interaction.

5.4.6 Quantitative treatment of ordinal predictor

Example 5.21 (Horseshoe crabs: color scores)

With scores $c = 1, 2, 3, 4$,

$$\operatorname{logit}\pi = \alpha + \beta_1 c + \beta_2 x, \qquad \hat\beta_1 = -0.509\ (0.224), \quad \hat\beta_2 = 0.458\ (0.104).$$

Each step darker multiplies the odds by $e^{-0.509} = 0.60$ at fixed width. Against (5.7), LR = 1.7 (df = 2, $P = 0.44$): the color parameters are consistent with a linear trend in the scores. The qualitative estimates (1.33, 1.40, 1.11, 0) suggest the binary scoring dark (0) versus other (1); against (5.7) it gives LR = 0.5 (df = 2), and the fit is

$$\operatorname{logit}\hat\pi = -12.980 + 1.300c + 0.478x \qquad (\widehat{\text{se}}:\ 0.526,\ 0.104). \tag{5.8}$$

At a given width, non-dark crabs have $e^{1.300} = 3.7$ times the odds of dark crabs. All three models say dark crabs are least likely to have satellites; deciding the best scoring needs more data. Treating an ordinal predictor quantitatively, when it fits, is simpler and gives more powerful one-df tests (Section 6.4).

5.4.7 Standardized and probability-based interpretations

To compare quantitative predictors with different units, one can fit with standardized predictors $(x_j - \bar{x}_j)/s_{x_j}$, or equivalently multiply $\hat{\beta}_j$ by s_{x_j}: the effect of a one-standard-deviation change. For a probability scale, Proposition 5.2 extends: at a setting with $\hat{\pi}$, a unit increase in x_j (others fixed) changes the probability by about $\hat{\beta}_j \hat{\pi}(1-\hat{\pi})$. More precisely, set the other predictors at their means and report $\hat{\pi}$ at the quartiles of x_j; for a dummy predictor report $\hat{\pi}$ at its two values.

Example 5.22 (Horseshoe crabs: effects on the probability scale)

For (5.8) at $\hat{\pi} = 0.5$, a 1-cm width increase raises π by about $0.478(0.5)(0.5) = 0.12$, considerable since 1 cm is under half a standard deviation. With $\bar{c} = 0.873$, $\hat{\pi} = 0.51$ at the lower width quartile 24.9 and 0.80 at the upper quartile 27.7. At $x = 26.3$, $\hat{\pi} = 0.40$ for dark and 0.71 for other colors.

Variable	Estimate	SE	Comparison	Change in probability
No interaction model				
Intercept	−12.980	2.727		
Color (0 dark, 1 other)	1.300	0.526	$(1, 0)$ at $\bar{x}$	$0.31 = 0.71 - 0.40$
Width x (cm)	0.478	0.104	(UQ, LQ) at $\bar{c}$	$0.29 = 0.80 - 0.51$
Interaction model				
Intercept	−5.854	6.694		
Color	−6.958	7.318		
Width x	0.200	0.262	(UQ, LQ) at $c = 0$	$0.13 = 0.43 - 0.30$
Width × color	0.322	0.286	(UQ, LQ) at $c = 1$	$0.29 = 0.84 - 0.55$

The width effect appears larger for lighter crabs, but the interaction is not significant.

5.5 Fitting logistic regression models

Motivation. Logistic regression is a canonical-link binomial GLM, so the theory of Sections 4.4–4.6 specializes neatly: the likelihood is concave, Newton–Raphson coincides with Fisher scoring and IRLS, and finite estimates exist exactly when the two outcome groups overlap.

Let $\mathbf{x}_i = (x_{i1}, \ldots, x_{ip})$ be setting i $(i = 1, \ldots, N)$, with $x_{i1} = 1$ for the intercept, and let y_i be the number of successes in n_i trials (with continuous predictors typically $N = n$ and $n_i = 1$). The model is

$$\pi(\mathbf{x}_i) = \frac{\exp\left(\sum_{j=1}^{p} \beta_j x_{ij}\right)}{1 + \exp\left(\sum_{j=1}^{p} \beta_j x_{ij}\right)}. \tag{5.9}$$

5.5.1 Likelihood equations

Theorem 5.23 (Logistic log likelihood and likelihood equations)

For independent $y_i \sim \text{bin}(n_i, \pi(\mathbf{x}_i))$ under (5.9),

$$L(\boldsymbol{\beta}) = \sum_j \Big(\sum_i y_i x_{ij}\Big)\beta_j - \sum_i n_i \log\Big[1 + \exp\Big(\sum_j \beta_j x_{ij}\Big)\Big], \tag{5.10}$$

so $\{\sum_i y_i x_{ij}\}$ are sufficient, and the likelihood equations are

$$\sum_i y_i x_{ij} - \sum_i n_i \hat{\pi}_i x_{ij} = 0, \quad j = 1, \dots, p, \qquad \text{i.e.} \qquad X^\top \mathbf{y} = X^\top \hat{\boldsymbol{\mu}}, \tag{5.11}$$

where $\hat{\mu}_i = n_i \hat{\pi}_i$.

Proof.

Step 1. The joint mass function is proportional to

$$\prod_i \pi_i^{y_i} (1-\pi_i)^{n_i - y_i} = \Big\{ \prod_i (1-\pi_i)^{n_i} \Big\} \exp\Big[\sum_i y_i \log \frac{\pi_i}{1-\pi_i} \Big].$$

Step 2. With $\operatorname{logit} \pi_i = \sum_j \beta_j x_{ij}$, the exponent is $\sum_i y_i \sum_j \beta_j x_{ij} = \sum_j \beta_j \sum_i y_i x_{ij}$, and $1 - \pi_i = [1 + \exp(\sum_j \beta_j x_{ij})]^{-1}$. Taking logs gives (5.10); the factorization theorem gives sufficiency.

Step 3. Differentiating,

$$\frac{\partial L}{\partial \beta_j} = \sum_i y_i x_{ij} - \sum_i n_i x_{ij} \frac{\exp(\sum_k \beta_k x_{ik})}{1 + \exp(\sum_k \beta_k x_{ik})} = \sum_i (y_i - n_i \pi_i) x_{ij}.$$

■

These are the binomial GLM equations (4.16) written with counts, an instance of Theorem 4.27: for canonical links the likelihood equations equate sufficient statistics to their estimated expected values (they are the normal equations of ordinary regression for the identity link). They are nonlinear and need iteration.

5.5.2 Asymptotic covariance matrix of parameter estimators

Theorem 5.24 (Information matrix for logistic regression)

$$-\frac{\partial^2 L}{\partial \beta_a \partial \beta_b} = \sum_i x_{ia} x_{ib}\, n_i \pi_i (1 - \pi_i),$$

which does not involve $\mathbf{y}$; so observed and expected information coincide, and

$$\widehat{\operatorname{Cov}}(\hat{\boldsymbol{\beta}}) = \{ X^\top \operatorname{diag}[n_i \hat{\pi}_i (1 - \hat{\pi}_i)] X \}^{-1}. \tag{5.12}$$

Proof. From Step 3 above, $\partial L / \partial \beta_a = \sum_i (y_i - n_i \pi_i) x_{ia}$, and by Proposition 5.2, Step 1, $\partial \pi_i / \partial \beta_b = \pi_i (1 - \pi_i) x_{ib}$. Hence $\partial^2 L / \partial \beta_a \partial \beta_b = -\sum_i n_i \pi_i (1 - \pi_i) x_{ia} x_{ib}$. This is free of y_i, so it equals its expectation, and Theorem 4.15 gives (5.12): the GLM form $(X^\top \hat{W} X)^{-1}$ with $w_i = n_i \pi_i (1 - \pi_i)$. ■

The square roots of the diagonal of (5.12) are the standard errors of $\hat{\boldsymbol{\beta}}$.

5.5.3 Distribution of probability estimators

For a setting $\mathbf{x}$, the fitted logit $\mathbf{x}^\top \hat{\boldsymbol{\beta}}$ has estimated variance $\mathbf{x}^\top \widehat{\operatorname{Cov}}(\hat{\boldsymbol{\beta}}) \mathbf{x}$, so

$$\mathbf{x}^\top \hat{\boldsymbol{\beta}} \pm z_{\alpha/2} \sqrt{\mathbf{x}^\top \widehat{\operatorname{Cov}}(\hat{\boldsymbol{\beta}}) \mathbf{x}}$$

is a large-sample interval for $\operatorname{logit} \pi(\mathbf{x})$; applying $t \mapsto e^t/(1+e^t)$ to the endpoints gives one for $\pi(\mathbf{x})$ (Proposition 5.6).

5.5.4 Newton–Raphson method applied to logistic regression

Proposition 5.25 (Newton–Raphson / IRLS for logistic regression)

With $\pi_i^{(t)} = \exp(\mathbf{x}_i^\top \boldsymbol{\beta}^{(t)})/[1+\exp(\mathbf{x}_i^\top \boldsymbol{\beta}^{(t)})]$ and $\mu_i^{(t)} = n_i \pi_i^{(t)}$, the Newton–Raphson step is

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} + \{X^\top \mathrm{diag}[n_i \pi_i^{(t)}(1-\pi_i^{(t)})] X\}^{-1} X^\top (\mathbf{y} - \boldsymbol{\mu}^{(t)}). \tag{5.13}$$

Equivalently $\boldsymbol{\beta}^{(t+1)} = (X^\top V_t^{-1} X)^{-1} X^\top V_t^{-1} \boldsymbol{z}^{(t)}$, with $V_t = \mathrm{diag}\{1/[n_i \pi_i^{(t)}(1-\pi_i^{(t)})]\}$ and

$$z_i^{(t)} = \log \frac{\pi_i^{(t)}}{1-\pi_i^{(t)}} + \frac{y_i - n_i \pi_i^{(t)}}{n_i \pi_i^{(t)}(1-\pi_i^{(t)})}. \tag{5.14}$$

At convergence, $-H^{-1}$ is the estimated covariance (5.12).

Proof. By Theorems 5.23 and 5.24, $\boldsymbol{u}^{(t)} = X^\top(\mathbf{y} - \boldsymbol{\mu}^{(t)})$ and $H^{(t)} = -X^\top \mathrm{diag}[n_i \pi_i^{(t)}(1-\pi_i^{(t)})] X$; substitute in (4.25). Since $H = -\mathcal{I}$ (canonical link), this is also Fisher scoring, hence IRLS by Theorem 4.26. For the logit, $\partial \eta_i / \partial \mu_i = 1/[n_i \pi_i (1-\pi_i)]$ when $\mu_i = n_i \pi_i$, so the working response (4.28) is (5.14), and $W = V_t^{-1}$. ■

The elements of V_t are the estimated asymptotic variances of sample logits (Section 3.1.6), and $\boldsymbol{z}^{(t)}$ is the linearized logit of the sample data. Starting from $\boldsymbol{\beta}^{(0)}$ (e.g. $\mathbf{0}$), the iterates converge to $\hat{\boldsymbol{\beta}}$ (Walker and Duncan 1967).

5.5.5 Convergence and existence of finite estimates

Theorem 5.26 (Concavity and uniqueness)

If X has full column rank, $L(\boldsymbol{\beta})$ in (5.10) is strictly concave; hence the ML estimate, when it exists, is unique, and any root of (5.11) is it.

Proof. By Theorem 5.24, $\boldsymbol{c}^\top H \boldsymbol{c} = -\sum_i n_i \pi_i (1-\pi_i)(\mathbf{x}_i^\top \boldsymbol{c})^2$ with every $n_i \pi_i (1-\pi_i) > 0$. It is < 0 unless $X\boldsymbol{c} = \mathbf{0}$, i.e. $\boldsymbol{c} = \mathbf{0}$. A strictly concave function has at most one maximizer, and a stationary point of a concave function is a global maximum. ■

Theorem 5.27 (Existence of the ML estimate: overlap versus separation)

Write the data as n individual binary observations $(\mathbf{x}_i, y_i)$ and let $s_i = 2y_i - 1 \in \{-1, 1\}$. Assume X has full column rank. Call the data *separated* if some $\boldsymbol{d} \neq \mathbf{0}$ satisfies $s_i \mathbf{x}_i^\top \boldsymbol{d} \geq 0$ for all i, with strict inequality for at least one i (a hyperplane with all $y = 1$ on one side and all $y = 0$ on the other, ties on it allowed). Otherwise the data *overlap*.
(a) If the data are separated, no ML estimate exists: L increases indefinitely along $\boldsymbol{d}$ and some $|\hat{\beta}_j|$ is "infinite".
(b) If the data overlap, a finite ML estimate exists, and it is unique.

Proof. [Supplement]

Step 1. (A convenient form.) For one binary observation, $y\eta - \log(1+e^\eta) = -\log(1+e^{-s\eta})$ (check $y = 1$ and $y = 0$). Hence

$$L(\boldsymbol{\beta}) = -\sum_i \log[1 + \exp(-s_i \mathbf{x}_i^\top \boldsymbol{\beta})] < 0.$$

Each term is a strictly increasing function of $s_i \mathbf{x}_i^\top \boldsymbol{\beta}$.

Step 2. (a) Suppose $\boldsymbol{d}$ separates and $\hat{\boldsymbol{\beta}}$ were a maximizer. For $t > 0$, $s_i \mathbf{x}_i^\top (\hat{\boldsymbol{\beta}} + t\boldsymbol{d}) \geq s_i \mathbf{x}_i^\top \hat{\boldsymbol{\beta}}$ for all i, with strict inequality for some i. By Step 1, $L(\hat{\boldsymbol{\beta}} + t\boldsymbol{d}) > L(\hat{\boldsymbol{\beta}})$, a contradiction. So the supremum

is not attained; along $\boldsymbol{\beta} + t\boldsymbol{d}$ the likelihood keeps increasing as $t \to \infty$ (under complete separation, $s_i \mathbf{x}_i^\top \boldsymbol{d} > 0$ for all i, it tends to the perfect-fit value $L = 0$).

Step 3. (b) Overlap and full rank imply: for every unit vector $\boldsymbol{d}$, $m(\boldsymbol{d}) = \min_i s_i \mathbf{x}_i^\top \boldsymbol{d} < 0$. Indeed, otherwise $s_i \mathbf{x}_i^\top \boldsymbol{d} \geq 0$ for all i; not separated forces all equalities, so $X\boldsymbol{d} = \mathbf{0}$, contradicting full rank.

Step 4. m is continuous on the compact unit sphere, so $\max_{\|\boldsymbol{d}\|=1} m(\boldsymbol{d}) = -c < 0$. For $\boldsymbol{\beta} = r\boldsymbol{d}$ with $r > 0$, $\|\boldsymbol{d}\| = 1$, choose i with $s_i \mathbf{x}_i^\top \boldsymbol{d} \leq -c$; then by Step 1

$$L(r\boldsymbol{d}) \leq -\log[1 + \exp(rc)] \leq -rc.$$

Step 5. Hence $\{\boldsymbol{\beta} : L(\boldsymbol{\beta}) \geq L(\mathbf{0})\} \subset \{\|\boldsymbol{\beta}\| \leq -L(\mathbf{0})/c\}$ is bounded; it is closed by continuity, hence compact and nonempty. The continuous L attains its maximum on it, which is a global maximum. Uniqueness is Theorem 5.26. ■

Similar results hold for the probit and some other links (Silvapulle 1981). In practice software rarely detects $\hat{\beta} = \pm\infty$: after a few cycles the log likelihood looks flat, convergence criteria are met, and enormous standard errors are reported.

Example 5.28 (Perfect discrimination)

Let $y = 0$ at $x = 10, 20, 30, 40$ and $y = 1$ at $x = 60, 70, 80, 90$. The ideal fit has $\hat{\pi} = 0$ for $x \leq 40$ and $\hat{\pi} = 1$ for $x \geq 60$. Taking $\alpha = -50\beta$ (so $\hat{\pi}(50) = 0.5$) and letting $\beta \to \infty$ gives ever larger likelihood approaching a perfect fit (Figure 2). PROC GENMOD reports $\operatorname{logit} \hat{\pi} = -192.2 + 3.8x$ with standard errors 8.0×10^8 and 1.5×10^7, numerical artifacts of $\hat{\beta} = \infty$.

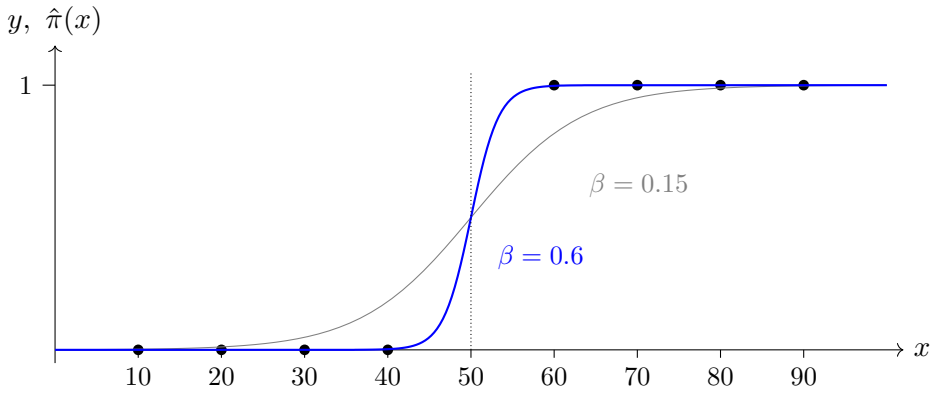


Figure 2: Complete separation: with $\alpha = -50\beta$ the fitted curve sharpens toward a step at $x = 50$ as $\beta \to \infty$, and the likelihood increases without attaining a maximum.